

Total No. of Questions :12]

P2962

SEAT No. :

[Total No. of Pages :3

[5154] - 514

B.E. (Civil)

DAMS AND HYDRAULIC STRUCTURES

(2012 Course) (Semester - II) (End Sem.)

Time : 2½ Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) *Answer any one from questions 1 or 2, 3 or 4, 5 or 6, 7 or 8, 9 or 10, 11 or 12.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data if necessary.*

Q1) a) State the objectives regarding the instrumentation in dam safety. **[3]**

b) Differentiate between large dam and small dam. **[3]**

OR

Q2) a) What are the types of arch dams? Write the merits and demerits of arch dams. **[3]**

b) What factors govern the selection of type of dam? **[3]**

Q3) a) State any three forces acting on gravity dam and write their equations. **[3]**

b) Write short note on: **[3]**

i) Constrution Joint.

ii) Drainage Gallery.

OR

Q4) a) Write a note on galleries in gravity dam with respect to location, size, shape and function. **[3]**

b) Write short note on buttress dams. **[3]**

P.T.O.

- Q5) a)** Draw neat sketch of any one type of spillway gate and explain. [4]
b) Write short note on safety and maintenance of spillway gate. [4]

OR

- Q6) a)** Explain pumped storage type of hydro-electric power plant. [4]
b) Write advantages and limitations of hydro power plant. [4]

- Q7) a)** Explain Swedish slip circle method of stability analysis with neat sketch. [8]
b) Write note on Khosla's theory application for design of structure on permeable foundations. Also explain the importance of exit gradient. [8]

OR

- Q8) a)** Explain Bilgh's theory of seepage with neat sketch. State its limitations. [8]
b) Determine the factor of safety of downstream slope of (homogenous section) an earthen dam drawn to a scale of 1:750 for the following data. [8]

- i) Area of N- rectangle = 20 cm^2
- ii) Area of T- rectangle = 10 cm^2
- iii) Length of slip circle arc = 20 cm^2
- iv) Angle of internal friction = 26°
- v) Cohesion $C' = 4000 \text{ Kg/m}^2$
- vi) Specific weight of soil = 1760 Kg/m^3

- Q9) a)** How irrigation canals are classified? Also describe the various considerations made in the alignment of an irrigation canal. [8]
b) Design an irrigation canal in alluvial soil according to lacey's factor theory. [8]

- i) Full supply discharge = $12 \text{ m}^3/\text{s}$,
- ii) Lacey's silt factor = 1,
- iii) Channel side slope = 1/2:1

OR

Q10)a) State various types of canal falls and explain any one with the help of neat sketch. [8]

b) Explain the procedure of designing cross regulator. [8]

Q11)a) Write short note on: [10]

- i) Launching Aprons,
- ii) Stepped fall,
- iii) Weir type escape,
- iv) Hokey head groynes,
- v) River training work.

b) What is Groynes? State the classification of Groynes. [8]

OR

Q12)a) Write short note on: [10]

- i) Pipe aqueduct,
- ii) Super passages,
- iii) Syphon aqueduct,
- iv) Level crossing,
- v) Inlet and outlet.

b) What do you mean by C.D. works? Write the factors for selection of C.D. Works. Also explain design considerations of it. [8]

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